

Module 5 — HIJING Hands-On

Week 5 · Build, run, and analyze Au+Au events

5.1 Source and build

The SPHENIX-patched mirror is the most reliable starting point (works with modern gfortran):

```
git clone https://github.com/SPHENIX-Collaboration/HIJING.git
cd HIJING
make # produces ./hijing and libhijing.a
./hijing # runs with stdin prompts, see below
```

5.2 Input parameters

HIJING accepts input either interactively on stdin or as a text file. Example input for 0–10% central Au+Au at $\sqrt{s_{NN}} = 200$ GeV (file hijing_aa.in):

```
hijing_aa.root ! output file (if compiled with ROOT hook)
200.0 ! sqrt(s_NN) in GeV
'A','A' ! projectile,target
197 79 197 79 ! A_proj Z_proj A_targ Z_targ
0.0 4.9 ! b_min b_max in fm
1000 ! number of events
```

Key physics switches (set via COMMON /HIPARNT/ HIPR1, IHPR2 in source, or via the Fun4All wrapper later):

- IHPR2(4) = 0/1 — jet quenching off/on
- IHPR2(3) = 0/1 — hard scattering off/on
- IHPR2(6) = 0/1 — shadowing off/on
- HIPR1(8) = 2.0 — mini-jet pT cutoff p_0 in GeV

5.3 Output format

Stock HIJING writes an ASCII event file in the HIJING-native format. Each event begins with a header (event number, N_{part} , b, ...) then one line per particle: STATUS PID PX PY PZ E M VX VY VZ. You will typically convert this to HepMC3 for downstream use. A minimal converter:

```
#include "HepMC3/GenEvent.h"
#include "HepMC3/WriterAscii.h"
#include "HepMC3/FourVector.h"
#include <fstream>
#include <sstream>
using namespace HepMC3;

int main(int argc, char** argv) {
    std::ifstream in(argv[1]);
    WriterAscii out(argv[2]);
    std::string line;
    while (std::getline(in, line)) {
        if (line.rfind("# EVENT",0) != 0) continue;
```

```
// ... (parse header, then read N particles)
GenEvent evt(Units::GEV, Units::MM);
// fill particles with FourVector(px,py,pz,E)
out.write_event(evt);
}
return 0;
}
```

Tip: The sPHENIX Fun4All module Fun4AllHepMCInputManager can also read the native HIJING output directly via its built-in adapter, saving you from writing a parser. This is what we use in Module 8.

5.4 Quick-look analysis macro

Once you have a ROOT or HepMC3 file, a $dN/d\eta$ plot is a good sanity check. For RHIC Au+Au 0–10%, expect $dN/d\eta|_{\eta=0} \approx 650$ for charged particles.

```
TH1F *hEta = new TH1F("hEta",";#eta;dN/d#eta",60,-6,6);
// loop over events, fill once per charged final particle with weight 1/nEvents/binWidth
hEta->Scale(1./nEvents/hEta->GetBinWidth(1));
hEta->Draw();
```

5.5 Exercises

- Generate 500 events in three b ranges (0–3, 6–8, 10–13 fm). Plot $dN/d\eta$ for each.
- Toggle IHP2(4) to compare hadron p_T spectra at $|\eta| < 1$ with quenching on vs. off.
- Parse the event header to extract N_{part} ; plot N_{part} vs. b .

5.6 Self-assessment

- ☐ HIJING builds and runs on my machine.
- ☐ I produced a $dN/d\eta$ plot consistent with RHIC published values ($\pm 10\%$).
- ☐ I understand at least three of the IHP2 switches.